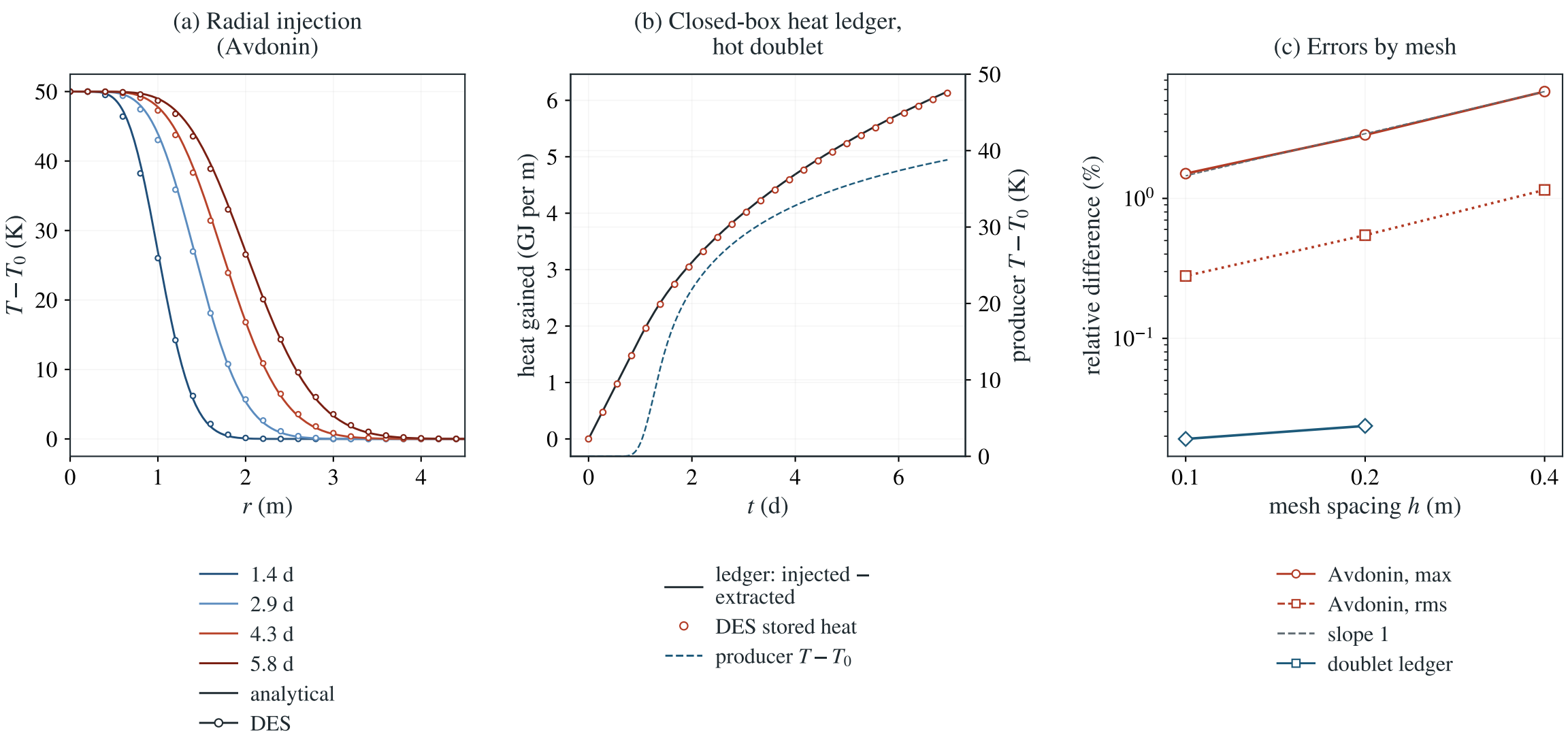

(a) Radial injection (Avdonin)
$T - T_0$ (K)
r (m)
1.4 d
2.9 d
4.3 d
5.8 d
analytical
DES
(b) Closed-box heat ledger, hot doublet
heat gained (GJ per m)
producer $T - T_0$ (K)
t (d)
ledger: injected – extracted
DES stored heat
producer $T - T_0$
(c) Errors by mesh
relative difference (%)
mesh spacing h (m)
Avdonin, max
Avdonin, rms
slope 1
doublet ledger